HETEROSCEDASTICITY

Introduction

An essential assumption of the classical linear regression model is that the disturbances u_i appearing in the population regression function are homoscedastic; that is, they all have the same variance. The word comes from the Greek verb skedanime, which means homoscedasticity, or equal (homo) spread (scedasticity), that is, equal variance.

Symbolically,

$$E(u_i^2) = \sigma^2 \qquad i = 1, 2, \ldots, n$$

Diagrammatic Representation of Homoscadesticity and Heteroscadesticity

To make the difference between homoscedasticity and heteroscedasticity clear, assume that in the two-variable model.

$$y_i = \beta_1 + \beta_2 x_i + u_i$$

y represents savings and x represents income.

Figures 1 and 2 show that, on average, savings increase as income rises. However, in Figure 1, the variance of savings remains constant at all income levels, whereas in Figure 2, it increases with income. It appears that, on average, higher-income families save more than lower-income families, but there is also greater variability in their savings.

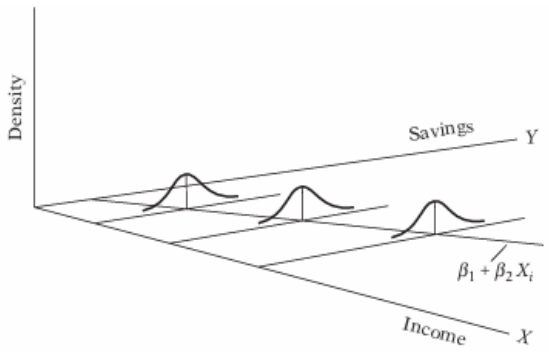


Figure 1. Homoscedasticity

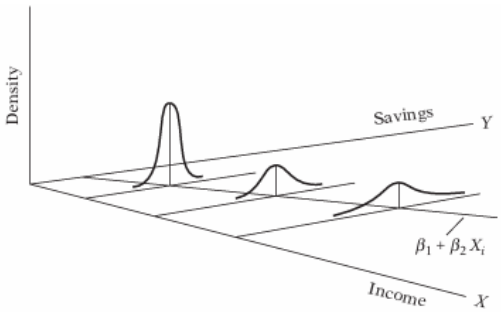


Figure 2. Heteroscedasticity

Interpretation of Figure 1.

The X-axis represents Income (independent variable).
The Y-axis represents Savings (dependent variable).
The Density axis shows the distribution of savings values for given income levels.

In the figure:

- Each distribution curve (bell-shaped) has equal spread (variance) along different points of the regression line $\beta_1 + \beta_2 x_i + u_i$
- This uniform spread across income levels signifies constant variance, a hallmark of homoscedasticity.

Example of Homoscedasticity

Consider a study analysing the relationship between household income (X) and monthly savings (Y).

- If families with different income levels exhibit similar variability in their savings patterns — for instance, regardless of income, savings deviate roughly the same from the predicted value—this reflects homoscedasticity.
- For example, if families earning ₹20,000, ₹50,000, and ₹1,00,000 all show similar fluctuations in their savings (e.g., ± ₹2,000 variation), the data follows a homoscedastic pattern.

Why is Homoscedasticity Important?

- Homoscedasticity ensures that the Ordinary Least Squares (OLS) estimators remain BLUE (Best Linear Unbiased Estimators).
- Violation of this assumption (i.e., heteroscedasticity) may lead to inefficient estimates and unreliable hypothesis tests.

Interpretation of Figure 2:

The X-axis represents Income (X), the independent variable.
The Y-axis represents Savings (Y), the dependent variable.
The curved lines on the graph depict the distribution of residuals (errors) at different income levels.
The variance of the errors (indicated by the width of the bell-shaped curves) changes as income increases, indicating non-constant variance, which is characteristic of heteroscedasticity.

Key Observation:

- At lower income levels, the spread of the residuals is relatively small, indicating low variance.
- As income increases, the spread of residuals becomes larger, suggesting higher variance.
- This pattern violates the assumption of homoscedasticity (constant variance of errors), a key requirement in Ordinary Least Squares (OLS) regression.

Solutions to Heteroscedasticity:

1. **Transformation of Variables:** Taking logarithms or square roots can stabilise variance.
2. **Weighted Least Squares (WLS):** Assigning weights inversely proportional to the variance at each observation.